

MBD4-LOF Dual Therapeutic Framework: Cytidine + ATRi

MBD4 Loss-of-Function



**BER Pathway Deficiency
(Glycosylase loss)**



**Unprocessed U:G Mismatches
at CpG Sites**

*BER substrate
accumulation*



**Accumulated U:G
Mismatch Substrate**



**Gemcitabine / Cytarabine
Exploit BER Substrate**



**AXIS 1: CYTIDINE ANALOGS
(Validated — isogenic + rescue)**

*Constitutive
fork stalling*



**Fork-Blocking Lesions
→ Replication Stress**



**ATR Checkpoint
Dependency**



**AXIS 2: ATR INHIBITION
(p=0.003, d=-0.74; 4/4 stress)**

*No PARP1 Upregulation
(No Selective Transcriptional Induction)*

 **REPLICATION FORK COLLAPSE**

